

ECE264-002 Spring 2020 Exam 3

Thursday, Apr 9th, 2020

1. Binary Search Tree

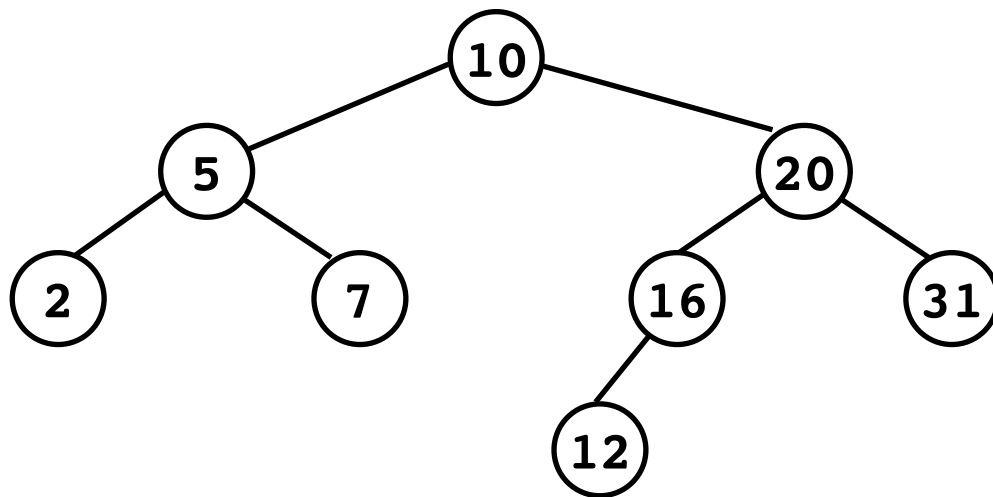
The following is the definition of node in a binary tree.

```
1.  typedef struct node {
2.      struct node * left;
3.      struct node * right;
4.      int value;
5.  } TreeNode;
```

The following is a function to insert node to a binary search tree.

```
1.  static TreeNode * _create_node(int val) {
2.      TreeNode * tn = malloc(sizeof(*tn));
3.      tn = {.left = NULL, .right = NULL, .value = val};
4.      return tn;
5.  }
6.
7.  TreeNode * insert_bst(TreeNode * tn, int val) {
8.      // if empty, create a node
9.      if (tn == NULL) {
10.         return _create_node(val);
11.     }
12.     // if not empty
13.     if (val == (tn->value)) {
14.         return tn;
15.     }
16.     // 1.A: fix the following code
17.     if (val 1.A (tn->value)) {
18.         tn->left = insert_bst(tn->left, val);
19.     }
20.     else {
21.         tn->right = insert_bst(tn->right, val);
22.     }
23.     return tn;
24. }
```

Consider the following binary search tree in question 1.B-1.E.



1.B: What is the **sixth** number in the **pre-order** traversal?

1.C: What is the **sixth** number in the **in-order** traversal?

1.D: What is the **sixth** number in the **post-order** traversal?

We use the above `insert_bst` function to insert node 17, 8, 7, 18, 35, 20, 6 (in this order) to the above binary search tree. After inserting the nodes, please answer question 1.E - 1.I

1.E: What is the **first** number in the **pre-order** traversal?

1.F: What is the **fifth** number in the **pre-order** traversal?

1.G: What is the **seventh** number in the **in-order** traversal?

1.H: What is the **last** number in the **post-order** traversal?

1.I: What is the **eighth** number in the **post-order** traversal?

Answer:

1.A: <

1.B: 16

1.C: 16

1.D: 31

1.E: 10

1.F: 6

1.G: 12

1.H: 10

1.I: 17

pre-order traversal: 10, 5, 2, 7, 20, 16, 12, 31

in-order traversal: 2, 5, 7, 10, 12, 16, 20, 31

post-order traversal: 2, 7, 5, 12, 16, 31, 20, 10

After insertion

pre-order traversal: 10, 5, 2, 7, 6, 8, 20, 16, 12, 17, 18, 31, 35

in-order traversal: 2, 5, 6, 7, 8, 10, 12, 16, 17, 18, 20, 31, 35

post-order traversal: 2, 6, 8, 7, 5, 12, 18, 17, 16, 35, 31, 20, 10

2. Huffman Coding

Consider a file that contains the following letters with the corresponding frequencies.

Letter	G	y	h	d	L	F	e
Frequency	1	2	8	9	12	30	40

Build the code book based on Huffman coding and answer the following question:

2.A: What is the binary code of letter e?

2.B: What letter does the binary code 1110 stands for?

2.C: What is **length** of the longest binary Huffman code?

2.D: How many bytes are needed to store the **data** in the Huffman compressed file?

Suppose the beginning of the **data** of this Huffman compressed file is shown below (output of `xxd` command)

```
da77 9ebe ...
```

Here ... means more to come. For your reference, the output of `xxd -b` command for the **data** of the same Huffman compressed file is shown below.

```
11011010 01110111 10011110 10111110 ...
```

Please write down the first, third, fifth, seventh, and ninth letters in the beginning of the original file for answers 2.E - 3.I.

2.J: Consider the frequencies in Table 1, if we change the frequency of letter d to 12 and keep the frequencies of all other letters unchanged in the file, will the structure of Huffman tree change?

2.K: Consider the frequencies in Table 1 (frequency of letter d is 9), if we change the frequency of letter h to 5 and keep the frequencies of all other letters unchanged in the file, will the structure of Huffman tree change?

Answer:

2.A: 0

2.B: d

2.C: 6

2.D: 29

2.E: L

2.F: F

2.G: d

2.H: y

2.I: h

2.J: Yes

2.K: Yes

e 0

F 10

L 110

d 1110

h 11111

G 111100

y 111101

beginning of data:

LLFedGyehe